

Particle-in-cell simulation of a cylindrical plasma mass separator based on Penning discharge with thermionic cathodes

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Abstract

This work is devoted to the modeling of a Penning discharge with a hot cathode, which is used to create a background plasma with a radial electric field in a plasma mass separator. We used a 2D3V electrostatic particle-in-cell plus Monte Carlo Collision (PIC/MCC) simulation model with a gyrokinetic approximation for magnetized electrons and a geometrical scaling scheme. The proposed model predicts the development of a rotating spoke structure, leading to fluctuations in the electric potential. The fluctuation amplitude and the averaged radial profile of the potential are in qualitative agreement with experiment. On the example of a model flow of silver and lead ions, we have shown that potential fluctuations can impair separation, leading to a partial overlap of ion deposition spots on a substrate. Each of the separated fractions contain about 11% impurities.

Keywords: PIC, particle-in-cell, reflex discharge, plasma mass separation

(Some figures may appear in colour only in the online journal)

1. Introduction

Plasma separation methods, despite a long history of development [1–4], have recently attracted much interest [5–11]. There are three main applications of plasma separation: reprocessing of spent nuclear fuel [12–14], reprocessing of radioactive waste [15, 16], and recycling of rare earth elements [8].

The plasma separation methods imply the transfer of the separated substance to the plasma state, the spatial separation of ions of different masses, and the deposition of the separated ions on a substrate. The separation can be based on various plasma phenomena, which have been considered in

detail in recent reviews [5–7]. In this work, we consider a separator with crossed electric and magnetic fields, the concept of which was proposed in [14] and practically implemented in the LaPlaS setup [17, 18]. The concept mainly implies for of spent nuclear fuel reprocessing. Such application requires separation of americium and curium (241–246 u) from lanthanides (139–169 u). For simulation of the ‘light’ and ‘heavy’ components of the compound, it was used silver (108 u) and lead (207 u) mixture.

The separator is a cylinder with an axial magnetic field, in which background plasma can be generated, in particular, by a Penning discharge. In such a configuration, a radial electric field arises in the background plasma. Separated ions are injected at some distance from the axis along the magnetic field. The radial field deflects the injected ions toward the axis, creating a velocity component perpendicular to the magnetic field.

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Ions with different mass-to-charge ratios are separated by an azimuthal angle.

Previously, using single-particle simulation, the optimal values of the magnetic and electric fields in the separator were found. The simulation makes it possible to remove the turning points of heavy and light ions as much as possible [19]. It has been experimentally shown that these parameters are achievable in the LaPlaS setup both in the configuration with a cold [20] and hot cathode [18]. Recently, the possibility of separation has been experimentally demonstrated for the first time [17, 18]. However, further interpretation of the experimental results and an increase in the separation efficiency require the use of numerical simulations that can reliably predict the physics of the Penning discharge and the self-consistent dynamics of the plasma flow of separated substances injected into it.

Despite the possibility of creating the required average radial electric field, the crossed fields in the Penning discharge cause instabilities [21–23], resulting in electric field fluctuations. This is a factor worsening separation [18]. To describe phenomena of this kind in discharges with crossed fields, the particle-in-cell (PIC) method is widely used, which makes it possible to realistically describe the development of instabilities and the spatiotemporal structure of the discharge [24–27].

In a cylindrical Penning discharge with a constant axial magnetic field, the plasma parameters do not depend on the position along the axis (with the exception of a narrow near-cathode region) [28], which makes it possible to simulate the discharge in a two-dimensional (azimuthal-radial) approximation. Previously, in this approximation, the formation of an azimuthally inhomogeneous spoke structure in a discharge rotating in the $E \times B$ direction was shown [29, 30]. It was established that the anomalous transport across the magnetic field occurs through the spoke [29], and the emerging spoke structure is apparently associated with the Simon–Hoh instability [30]. Alternative mechanism implies the appearance of the ionization waves [31].

In the cited works, the experimental setup described in [32, 33] is modeled, which is qualitatively similar to the LaPlaS setup. However, despite that the conditions for the instabilities development ($\vec{E} \nabla n > 0$) are still preserved, at the same time the physical parameters differ significantly. The LaPlaS separator has a larger size (diameter 50 cm instead of 26 cm in experiment [32] and 5 cm in the simulation [30]) and an order of magnitude higher magnetic field (1.4 kG instead 0.1 kG in previous study), which leads to magnetization of not only the electronic, but also the ionic component. The average radial electric field is also larger (10–15 V cm^{−1} instead of 2 V cm^{−1}). All these differences may lead to qualitatively different discharge physics.

In this work, the Penning discharge with a hot cathode is studied using the PIC method in the 2D3V approximation. The scaling rules are applied to simulate a scaled-down LaPlaS setup. The main task is to study both the averaged and time-resolved characteristics of the discharge, as well as to analyze the effect of azimuthal inhomogeneities on the quality of ion separation in the considered type of separator.

2. Computational details

A detailed description of the experimental setup with a thermionic cathode is given in [18]. Here, we briefly describe the simplified setup and the parameters of the simulated experiment.

The separator has the shape of a cylinder (figure 1). The magnetic field is directed along the separator axis. The side cylindrical wall acts as an anode and has a zero potential. The end cathodes have a potential of about −500 V and consist of two parts. The inner part is a thermionic cathode, which injects thermionic electrons into the separator. The cold part of the cathode has the shape of a ring. The ions of the model substances to be separated (Pb and Ag) are injected along the magnetic field, but with some displacement from the separator axis.

Using the PIC method, the cross section of the separator was modeled in 2D geometry. The computational domain was a circle with a diameter of $2R_0 = 50$ cm, at the boundary of which the zero potential was set. Thermoelectrons with a total current of 6 A, close to the current in experiment [34], were injected into the central region 2 cm in diameter. The initial energy of the electrons was 100 eV. This roughly corresponds to the cathodic potential drop in the experiment [17, 18, 34], due to which, as was assumed, the electrons acquire the given energy.

The total current through the cold part of the cathode (an annular region with a diameter of 2–30 cm) was set as 4 A [34]. This value includes the current flowing through the cathodes on both sides. About 2 A on each side. It was assumed that the current through the cold cathode is formed mainly by Ar⁺ ions drawn from the discharge by the cathodic potential drop. We took into account that ions neutralized at the cathode cause secondary electron emission with an emission coefficient of ~ 0.1 [35]. Thus, in the cross section of the cold cathode, the ion death rate was set to 3.6 A and the electron injection rate due to secondary emission was 0.4 A. Parameters of the simulated domain are listed in the table 1.

The death of electrons occurs at the anode. The cathode potential was not explicitly specified in the calculation. Neutral Ar atoms were not modeled explicitly. It was assumed that the gas is isotropically distributed and has a temperature of 500 K. The following elementary processes were taken into account: elastic collisions of electrons with neutrals, elastic collisions of ions with neutrals, ion-neutral charge exchange, ionization by electron impact. The cross sections of the processes are taken from [36, 37]. The Monte Carlo method was used to simulate elementary processes. Models of elastic collisions and ionization are taken from [38]. The simulation was performed in the electrostatic approximation. Neutral particle excitation was not modeled.

To make the calculation technically possible, the scaling of the simulated system by 50 times was used in accordance with the laws proposed in the work [39]. In table 2 there is a description of the scaling laws used with scaling coefficient $\zeta = 1/50$.

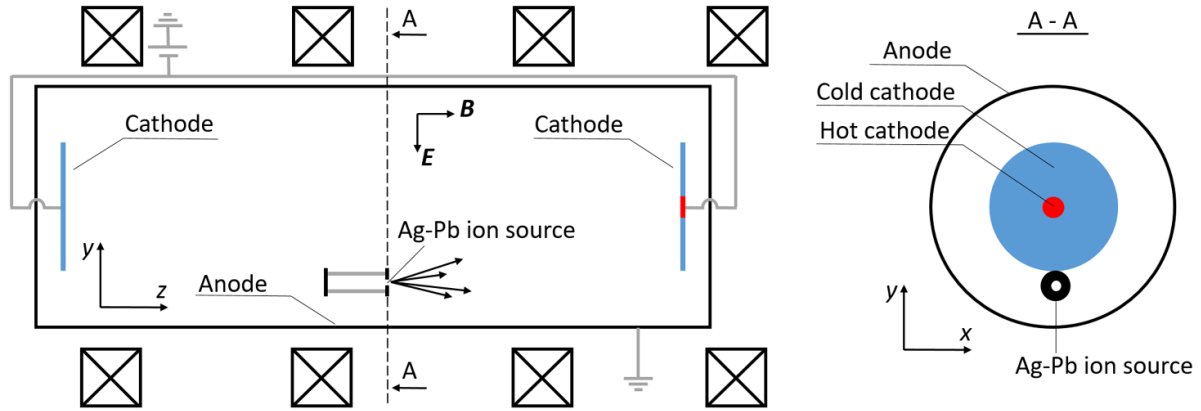


Figure 1. Scheme of the simulated setup.

Table 1. Parameters of simulated domain.

Separator diameter	50 cm
Thermionic cathode diameter	2 cm
Cold cathode diameter	30 cm
Magnetic field	0.1 T
Electron average energy	2 eV
Discharge current	10 A
Hot cathode current	6 A
Cold cathode current	4 A
Neutral temperature	500 K
Neutral pressure	4 mTorr
Electron, ion density	$\sim 10^{11} \text{ cm}^{-3}$

Table 2. Scaling laws [39].

Quantities	Real system	Scaled system
Domain length	L	$L\zeta$
Neutral density	n	$n\zeta^{-1}$
Discharge current	I	$I\zeta$
Magnetic field	B	$B\zeta^{-1}$
Electric potential	U	U

The Boris scheme [40] was used to integrate the equations of ion motion. To increase the time step, the magnetized electrons were modeled within the framework of the gyrokinetic approximation. The equation for the gyrokinetic center, the finite-difference scheme, and other details are given in [41]. Despite these accelerating approaches, the use of the real mass of argon makes the calculations still cumbersome. Therefore, the mass of the ion was reduced to $500m_e$.

The calculations were performed on a square grid of 70×70 cells. The Poisson equation was solved by the successive overrelaxation method [42]. The time step was 10 ps. The simulation was carried out using a self-made code PICSIMPL [43]. The parameters of the simulated scaled system are listed in table 3.

Table 3. Parameters of scaled system.

Scaling coefficient	$\zeta = 1/50$
Separator diameter	1 cm
Thermionic cathode diameter	0.04 cm
Cold cathode diameter	0.6 cm
Magnetic field	5 T
Hot cathode current	0.12 A
Cold cathode current	0.08 A
Neutral pressure	200 mTorr
Neutral temperature	500 K
Grid size	70×70 cells
Cell size, cm	1.43×10^{-2}
Electron temperature	2.1 eV
Electron emission energy	100 eV
Ion mass	$500m_e$
dt	10 ps
Reactions	$e + \text{Ar} \rightarrow e + \text{Ar}$ (elastic) $e + \text{Ar} \rightarrow 2e + \text{Ar}^+$ $\text{Ar}^+ + \text{Ar} \rightarrow \text{Ar}^+ + \text{Ar}$ (elastic) $\text{Ar}^+ + \text{Ar} \rightarrow \text{Ar} + \text{Ar}^+$

3. Results and discussion

3.1. Simulation of Penning discharge with thermionic cathode

The discharge was analyzed after the system reached a steady state. We judged the onset of the steady state by reaching a plateau in the total number of electrons and ions in the system. For the simulated system, the equilibration period was about $8 \mu\text{s}$ (figure 2).

Figure 3 shows the distribution of ions and electrons in the cross section of the separator at different times. It can be seen that the distribution is not uniform. In the central region, the density is higher due to electrons emitted by the thermionic cathode. Accelerated by the cathode voltage drop to energies of ~ 100 eV, these electrons effectively ionize the background gas. When moving away from the central region, the plasma density decreases on average. At the same time, the plasma is also inhomogeneous in the azimuthal direction. A single-mode

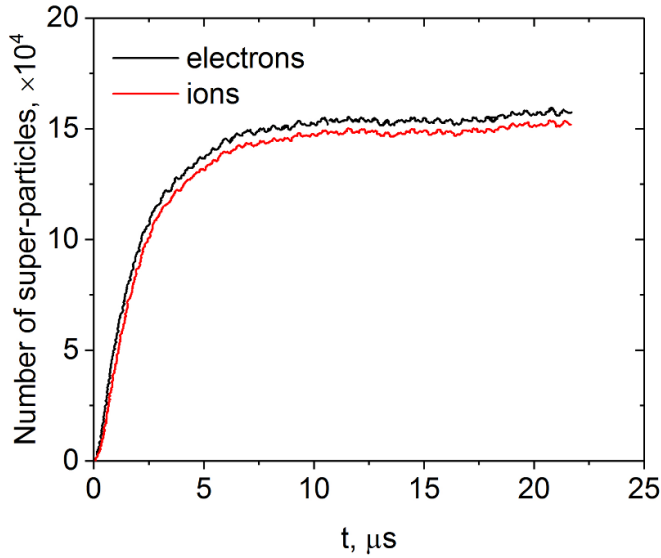


Figure 2. The number of super-particles as a function of time.

spoke structure is formed, rotating in the $E \times B$ direction (counterclockwise in figure 3). The spoke structure has a fairly distinct front with an increased density. The density decreases with distance from the front in a clockwise direction. It can be noted that the spoke structure is more delocalized in the azimuthal angle than in [30]. The contours of the increased density of ions and electrons at each moment of time coincide due to the electroneutrality of the plasma. The total turn this structure makes during $0.64 \mu\text{s}$. Thus the frequency can be estimated as 1.6 MHz. Assuming linear scaling of the time one can obtain frequency of 30 kHz, that quiet good agreement with the experiment [44].

Figures 4(a) and (b) show the temporal fluctuations of the electron and ion densities, as well as the electric potential at the point $(R_0/2; 0)$. It can be seen that the plasma density and potential fluctuate. Significant fluctuations can be distinguished, which corresponds to the rotation period of the spoke structure. At the moment when the front of the structure reaches the observation point, the density sharply increases from $\sim 1 \times 10^{17} \text{ m}^{-3}$ to $\sim 5 \times 10^{17} \text{ m}^{-3}$, after which a gentler decrease is observed. The spikes in plasma density are preceded by a potential drop from about -400 V to about -500 V with a time advance of the order of 50–100 ns (figure 4(b)). In addition to pronounced fluctuations with a period of $\sim 0.64 \mu\text{s}$, higher-frequency fluctuations associated with the turbulent nature of the plasma are observed. The time dependences of the electron and ion densities do not coincide exactly, which indicates a violation of local electrical neutrality.

Figure 4(c) shows the dependence of the density of ions and electrons at the point $(0.99R_0; 0)$, which is located in the vicinity of the anode. Figure 4(d) shows the time dependences of the current flowing through the anode segments, having an angle $\theta = \pi/2$ and $\theta = \pi$, normalized to the total anode current $I_{2\pi}$. The position of the anode segments with respect to the point $(0.99R_0; 0)$ is shown in figure 4(e). The current spikes are synchronized with the rotation of the spoke structure and indicate a nonuniform current distribution over

the anode surface. At the moment when the spoke maximally overlaps with the selected anode region, the current is maximal. This indicates that the electron transport through the spoke is intensified, which agrees with previous calculations [29, 30]. However, when the spoke leaves the selected region of the anode, the current does not drop to zero.

Figure 5 shows the radial potential profile averaged over $5 \mu\text{s}$ in comparison with the experiment [44]. It can be seen that the calculation well reproduces the linear dependence of the potential on the radius, while the value of the average radial electric field is overestimated in the calculation by about two times, which is a good agreement, taking into account the approximations and simplifications made in the model. The formation of a potential well on the axis is explained by the low mobility of magnetized electrons across the magnetic field compared to ions. Both experimental and calculated amplitude of potential fluctuations, which was estimated as a standard deviation are in the range of 40–30 V at $r < 20 \text{ cm}$.

As noted in the computational details, in addition to scaling the system, to speed up the calculations, the ion mass was reduced to $500m_e$, which may change the discharge parameters [39, 45]. To clarify this issue, we studied the influence of the ion mass on the averaged radial profile (figure 6(a)) and the potential well depth on the axis (figure 6(b)). It can be seen that the well depth decreases with increasing mass. This explains the overestimation of the average radial field in comparison with the experiment in figure 5. However, it can be seen from figure 6(b) that the rate of change in the well depth decreases with increasing mass, and for ion masses of the order of $500m_e$ and more, apparently, one can speak of a reasonable compromise between the calculation speed and accuracy.

3.2. Separation of lead and silver ions in the Penning discharge with a thermionic cathode

By analogy with the experiments [17, 18, 46], we studied the separation of model substances Ag and Pb, which imitate transuranium elements and fission products, respectively. The calculations were performed in the single-particle approximation. I.e. the Ag^+ and Pb^+ ions did not interact with each other and did not affect the plasma parameters in the separator. We performed the calculation of Ag^+ and Pb^+ trajectories in 3D-geometry. The azimuthal and radial potential were taken from 2D-3V model described in previous chapter 3.1. Potential along z -axis was set constant (electric field was zero along z -direction). Besides, in reality, the injected plasma flow can cause local perturbations of the background plasma potential in the injection region and affect the ion trajectories [18]. A more sophisticated self-consistent modeling of the plasma flow propagation in the separator is planned in the next work. A feature of our present calculation is that it takes into account the spatial structure of plasma potential fluctuations. In contrast to [18], where potential fluctuations were modeled in phase over the entire space, in this work, fluctuations have a pronounced inhomogeneity both in the azimuthal and radial directions. Thus, azimuthal electric fields appear which also affect the trajectories of the separating ions.

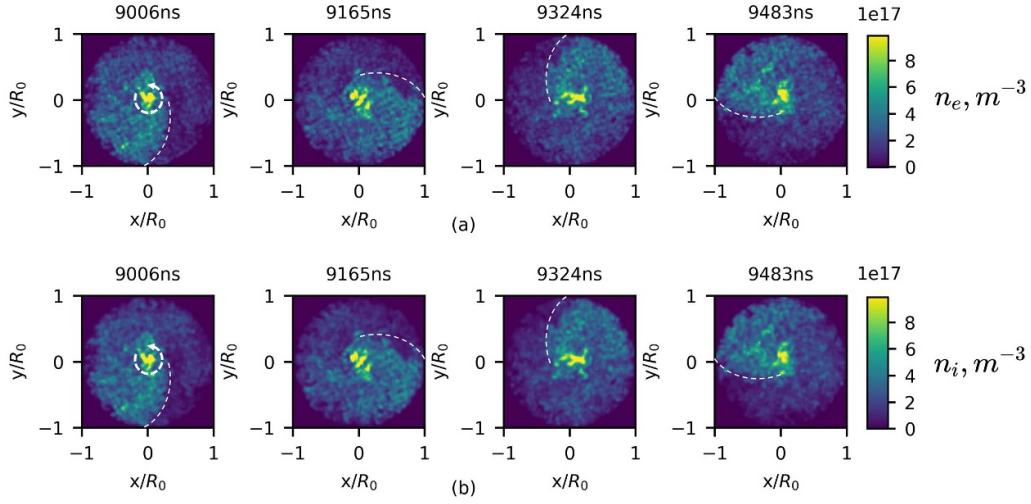


Figure 3. Time evolution of the discharge structure. Density distribution of electrons (a) and ions (b) at time points 9006 ns, 9165 ns, 9324 ns, 9483 ns. The counterclockwise rotating high-density plasma front is marked by a dotted line.

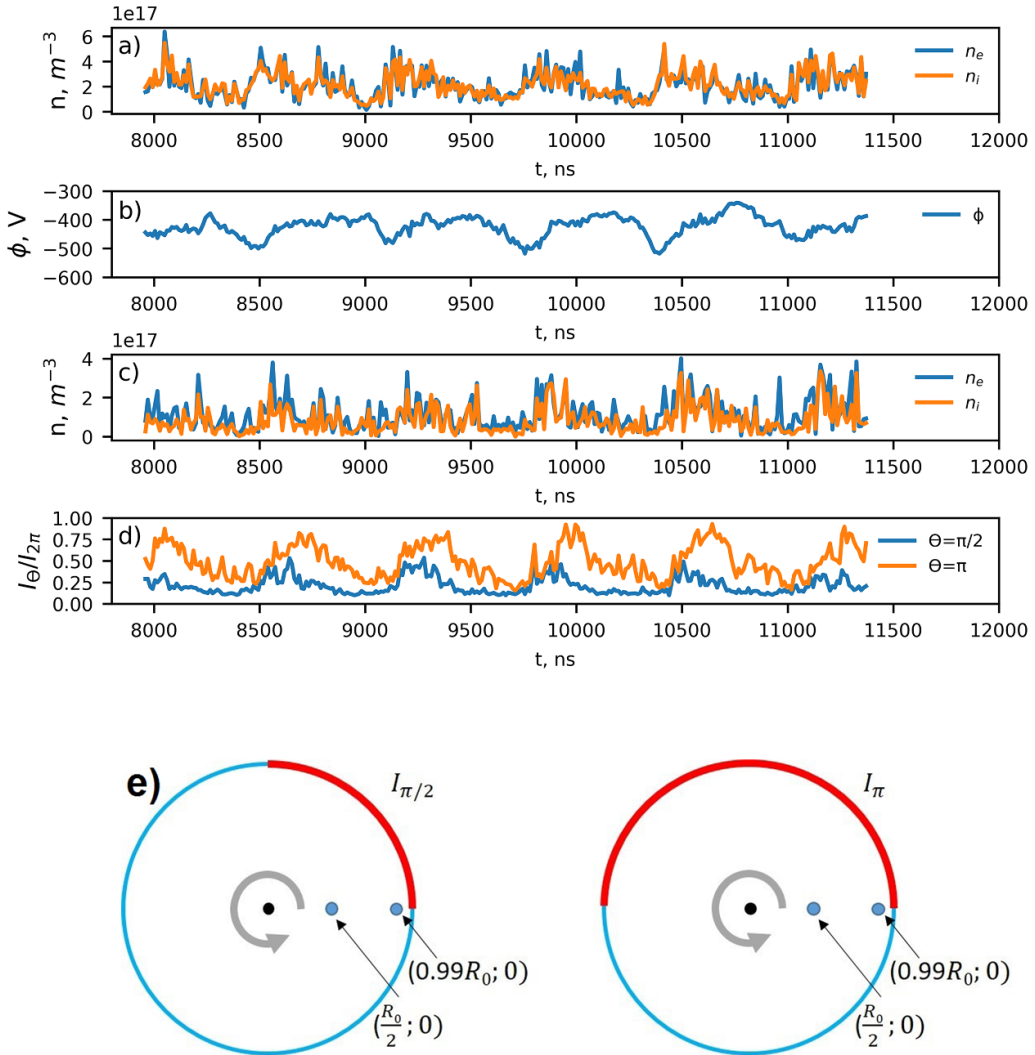


Figure 4. The time dependence of (a) the density of electrons and ions at the point $(R_0/2; 0)$; (b) the potential at the point $(R_0/2; 0)$; (c) the density of ions and electrons at the point $(0.99R_0; 0)$, which is located in the vicinity of the anode; (d) the current I_θ flowing through the anode segment with an angle θ , normalized to the total anode current $I_{2\pi}$; (e) the position of the anode segment with respect to the points $(R_0/2; 0)$ and $(0.99R_0; 0)$.

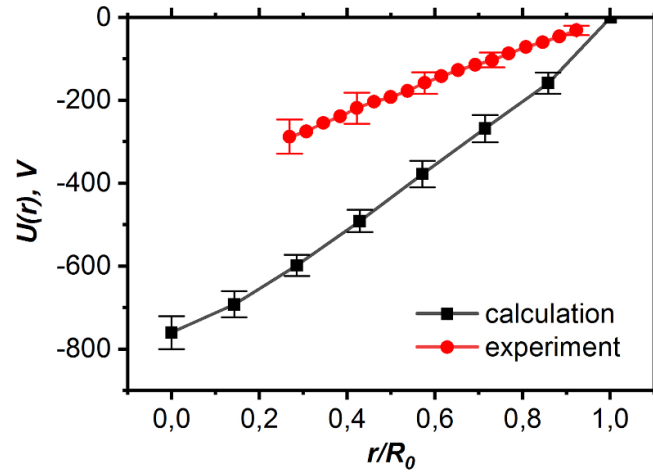


Figure 5. The averaged radial potential profile.

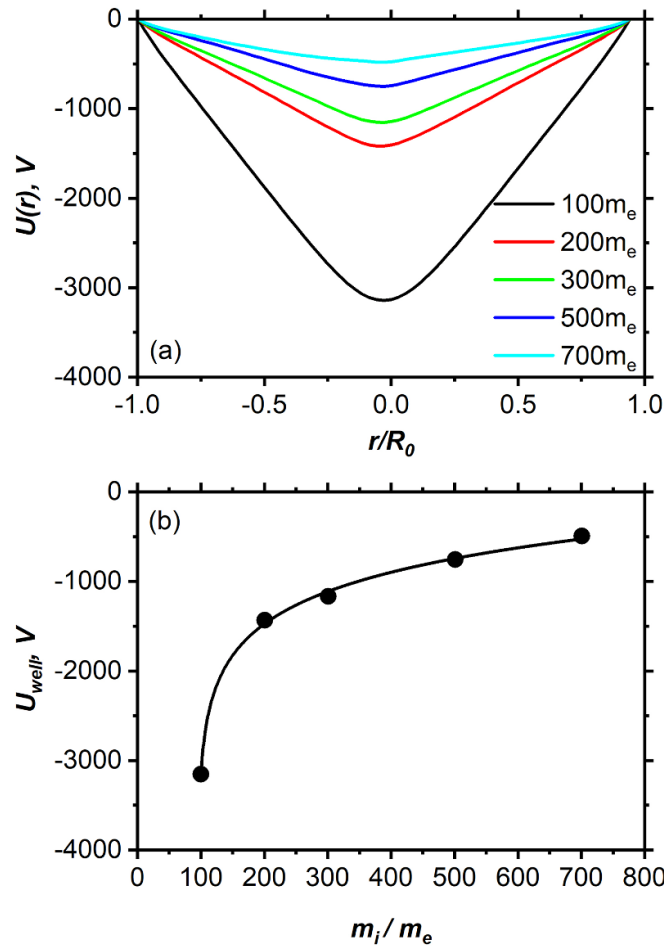


Figure 6. (a) The effect of ion mass on the radial potential profile in the separator; (b) the potential well depth on the axis as a function of ion mass.

Figure 7(a) shows the trajectories of Ag^+ and Pb^+ ions (in projection onto the cross section of the separator) propagating in the calculated average potential from figure 5. The ions were injected from the point (0.18; 0) along the magnetic field with an angular spread of 30° (see figure 1). The energy range of the ions was from 1 eV to 20 eV. The injected ions begin to deflect

towards the separator axis under the action of a radial electric field. However, they do not reach the potential minimum on the axis due to the magnetic force. Having passed the point of closest approach to the axis, the ions begin to slow down, because move in the direction of increasing potential, and stop at the turning point. Then the process is repeated. It can be seen

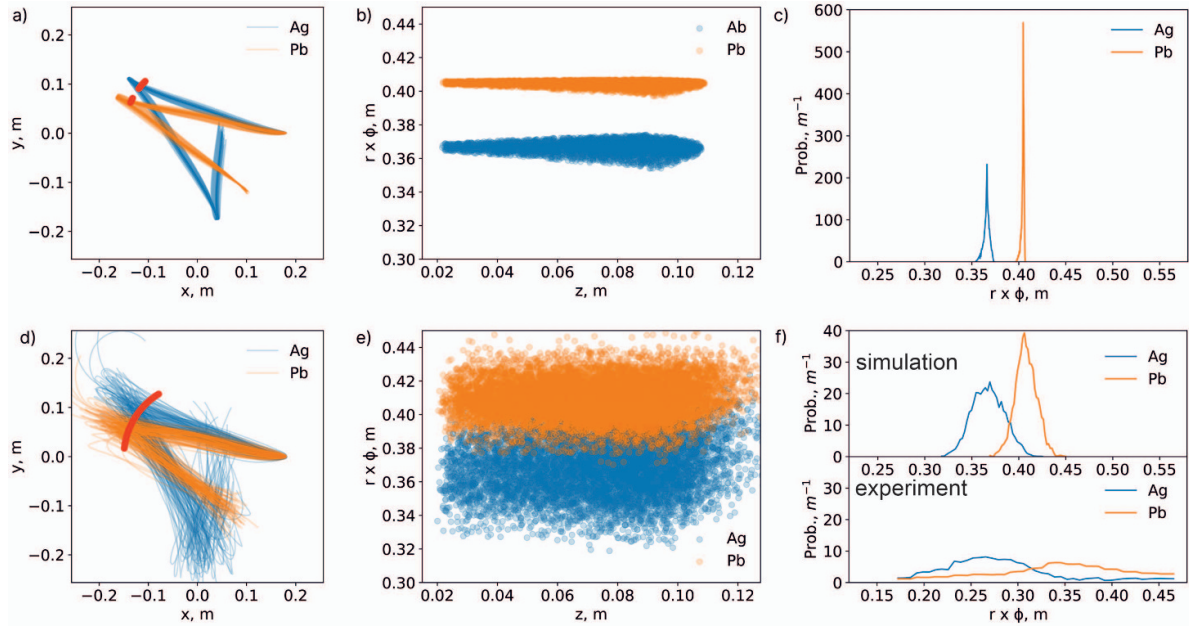


Figure 7. The two sets of the simulation results: in the average cylindrically symmetric potential shown in figure 5(a) (a) the trajectories of the Ag^+ and Pb^+ ions, (b) the deposition spots of Ag^+ and Pb^+ ions on a cylindrical collector with a radius of 15 cm, (c) distribution of ions on the collector, averaged along the z direction; in the potential with spatiotemporal fluctuations (d) the trajectories of Ag^+ and Pb^+ ions, (e) the deposition spots of Ag^+ and Pb^+ ions on a cylindrical collector with a radius of 15 cm; (f) distribution of ions on the collector, averaged along the z direction.

that ions of different masses are separated. Figure 7(b) shows the ion deposition spots on the surface of a cylindrical collector with a radius of 15 cm (shown by the red arc in figure 7(a)). Model substances are completely separated when moving in the average potential. Figure 7(c) shows the distribution of deposited ions averaged along the z direction.

Figure 7(d) shows the ion trajectories taking into account the spatiotemporal potential fluctuations obtained in the simulation. The separating flows become less focused. The deposition spots begin to overlap (figures 7(e) and (f)). Complete separation becomes impossible. If the substrate is cut along the intersection point of the distributions of deposited Pb and Ag in figure 7(f), each of the separated fractions will contain about 11% impurities in terms of the number of atoms.

Figure 7(f) also shows the experimental distributions from [47]. The distance between the maxima of the distributions in the experiment and calculation are in good agreement. In accordance with our previous work [19], a slightly smaller distance in the calculation can be explained by an overestimated value of the average radial electric field. For the same reason, the positions of the experimental maxima are shifted relative to the calculated ones by about 10 cm towards lower values of the $r \times \varphi$ coordinate. Experimental distributions are wider. Both components are detected in any area of the collector. This is probably due to the fact that the injected flow in the experiment, in addition to ionized atoms, contains neutral Pb and Ag atoms, whose motion is not related to the electromagnetic field. The processes of erosion and sputtering of the deposited substances are also possible, which is not taken into account in the calculation.

4. Conclusions

Using the PIC method, the spatiotemporal structure of the Penning discharge with a hot cathode, which is used to create a background plasma and a radial electric field in the LaPlaS setup, has been studied. The simulation was performed in 2D3V approximation using the scaling rules and the gyrokinetic approximation for magnetized electrons. It is shown that a rotating spoke structure is formed, which leads to fluctuations of the plasma parameters. At half the radius, the standard deviation of the potential is about 30 V, and the plasma density changes by a factor of ~ 5 . The radial potential profile and the oscillation amplitude are in qualitative agreement with experiment. It is shown that fluctuations with standard deviation of 30–40 V arising due to the instability in the discharge and gyrated in azimuthal direction lead to decrease of the separation coefficient. The separated lead, which mimics the actinides in this calculation, contains 11% silver impurities by the number of atoms. To increase the degree of separation, further efforts need to be made to reduce the influence of plasma potential fluctuations on the separated particles. One possible option is to create a background plasma with other types of sources (ICP [48], ECR [49] et al). Another way is to change the shape of the spatial distribution of plasma potential, so as to reduce the sensitivity of the separated particle trajectories to fluctuations. For example, in the [9], it has been shown that the most sensitive area of the trajectories are the pericentral regions. This is part of the trajectory closest to the axis of the system. By reducing the length of the trajectory near the pericenter, the influence of fluctuations will be reduced. In addition, more stringent conditions on the initial beam parameters (angle and

energy spread) will also lead to an increase in the degree of separation.

Data availability statements

All data that support the findings of this study are included within the article (and any supplementary files).

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